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## **Accelerating Field Interventions And Value Realization Through Intelligent Well Surveillance And Diagnostics**

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### **Abstract**

Acquiring and interpreting real-time data from both surface facilities and subsurface environments provides an accurate representation of wellbore and reservoir conditions. By capturing high-resolution and high-frequency environmental measurements, sensors enable continuous surveillance of the entire production system from reservoir inflow to surface processing. High and low frequency production data are ingested into digital engineering workflows that incorporate physics laws, analytics and domain expertise.

Despite significant advancements in data acquisition and analytics, many digital oil fields (DOF) continue to operate engineering workflows in functional silos, with surface production, reservoir management, and equipment diagnostics often managed independently. While these compartmentalized workflows are individually robust and capable of generating valuable insights within their domains, the lack of integration limits the holistic understanding of the production system. This fragmentation constrains the ability to fully integrate cross-disciplinary data insights, thereby limiting opportunities for comprehensive production optimization and proactive operational decision-making across the asset lifecycle. Addressing these silos is fundamental to realizing the full value of integrated DOF.

This paper presents the design and large-scale deployment of a pioneering, engineering-driven feedback loop platform for Well Surveillance and Diagnostics, developed under ADNOC's Intelligent Production System Optimization (AiPSO) initiative. Engineered to support high-well-count operations across both greenfield and brownfield assets, the platform tackles critical challenges including delayed issue detection, alarm fatigue, inconsistent diagnostics, and fragmented decision-making.

Integrated and orchestrated through an innovative framework, the system synchronizes inputs and outputs leveraging engineering expertise alongside automated data and physics model analysis to continuously validate insights, prioritize exceptions, and guide decision-making by embedding human-centered logic within a feedback loop that incorporates modular workflows covering virtual flow meters (VFM) for rate estimation, lift performance diagnostics, and smart ticketing to quantify added value, all aimed at accelerating cross-disciplinary collaboration and action-field response.

The deployment of this pioneering production ecosystem marks a shift away from the fragmented, siloed workflows that have long limited the full potential of DOF. By creating an innovative, engineering-driven feedback loop, the system successfully integrates data and expertise to provide a holistic view of the production system. The result is a transformative leap from passive monitoring to a proactive engine of production optimization. This integrated approach not only accelerates decision-making and cross-disciplinary collaboration but also quantifies value, demonstrating that a unified and intelligent framework is the key to unlocking true operational excellence and maximizing asset performance across the entire asset lifecycle.

## Introduction

The origin of fundamental well surveillance and production monitoring can be traced back to the earliest days of modern oil and gas extraction, when the monitoring of flowing wellhead pressure was a primary, and often the only, method of assessing a well health (K.B. Nowels 1932). Significant advancements in instrumentation have revolutionized modern well surveillance, moving far beyond the simple pressure gauges of the past. Today, a new generation of smart sensors and sophisticated devices provides a continuous, high-resolution stream of data from both the surface and downhole environments. Technologies like fiber-optic sensing, and permanent downhole gauges now measure a wide range of parameters in real-time, including temperature, pressure, flow rates, and even the chemical composition of fluids. These devices are often integrated with the Internet of Things (IoT), enabling wireless communication from remote locations and feeding data directly into advanced analytics platforms (Salim et al, 2023).

Nowadays, managing a high volume of wells and operations requires meticulous attention to detail to ensure optimal performance and output (Jatmiko et al, 2025, Mongensen et al, 2024). Given the scale and complexity of these interconnected systems, decision-making processes for production optimization and surveillance become inherently time-consuming. The vast number of wells and the intricate network of pipelines necessitate a thorough analysis, evaluation, and strategic planning before implementing any changes or improvements. Effective well surveillance is a cornerstone of production optimization in mature oilfields. As well counts increase and artificial lift systems diversify, traditional surveillance approaches face major limitations. Manual analysis of alarms, inconsistent diagnostics, and delayed interventions often result in deferred production and increased downtime.

## From Data Acquisition to Visual Analytics

In modern digital oil field systems Data, Workflows, Domain Expertise, and Visualization form an ecosystem that drives informed decision-making and operational optimization. **Data** that can be structured, unstructured, or real-time sensor streams and serves as the foundational input, capturing the state of systems and processes. **Workflows** provide logical and automated sequences of operations that transform, validate, and route this data through analytical models, simulation tools, or decision-support algorithms (Al Balushi et al, 2013). **Domain expertise** embeds human knowledge and reservoir and production domain experience, contextual understanding, and operational constraints into these workflows, ensuring that the processes remain aligned with real field operational conditions. Finally, **Visualization** translates complex, multi-dimensional outputs into intuitive, interactive representations (dashboards), enabling engineers to rapidly interpret trends, anomalies, and insights.

## From Siloed Engineering Workflows to innovative feedback loop ecosystem

Engineering workflows that operate in silos, where teams, departments and engineers work in isolation with limited communication, data sharing, and cross-functional collaboration can severely hinder decision-making. As a result, decisions are frequently based on incomplete or outdated information, increasing operational risk, delaying corrective actions, and reducing optimization potential.

Overcoming these challenges requires a unified and integrated workflow ecosystem, where data, models, domain expertise and visual analytics are seamlessly connected, enabling multidisciplinary teams to operate with shared context, enhanced situational awareness, and the agility to respond to dynamic field conditions.

**Feedback loop.** At its simplest, a feedback loop is a process where a systems output or some other controlled variable is routed back into the system as an input or an input to some other internally situated component or subsystem), influencing future outputs. (Distefano et al, 1990). An orchestrated multi-workflow feedback loop is a coordinated control architecture in which several domain-specific workflows each with its own localized sensing, decision, and actuation cycle are integrated via a shared data and execution framework to operate as a unified system (Figure 1). Each workflow loop processes real-time inputs, generates outputs, and passes these outputs as structured inputs or control constraints to other loops, creating interdependencies that span multiple operational domains and time scales.

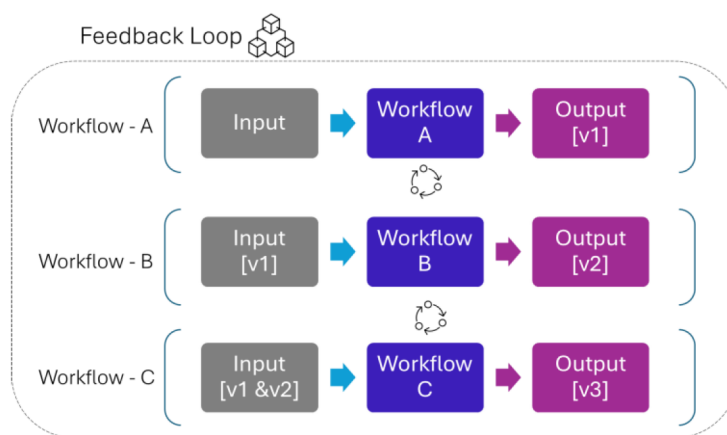


Figure 1—Integrated Workflows Feedback loop

## Data Architecture

The production ecosystem is described in a complete orchestration diagram (Figure 2) from source to visualization; it begins with data acquisition, where raw data is captured from various sources such as sensors, or historical databases. This incoming data undergoes ingestion through batch pipelines or streaming frameworks into a staging environment. Next, data processing and transformation steps standardize formats, perform cleansing, apply domain-specific rules (Abdul Aziz et al, 2008). The transformed datasets are then stored in optimized storage layers data warehouses for analytical queries or data lakes for unstructured and semi-structured content. Semantic modeling and aggregation layers are applied to align data with business or operational metrics. Finally, visualization platforms connect to these curated datasets, rendering interactive dashboards and dynamic visualizations that enable users to explore trends, alarms and insights with performance often enhanced through caching and pre-computed views. This end-to-end pipeline is orchestrated by workflow management systems to ensure reliability, scalability, and traceability.

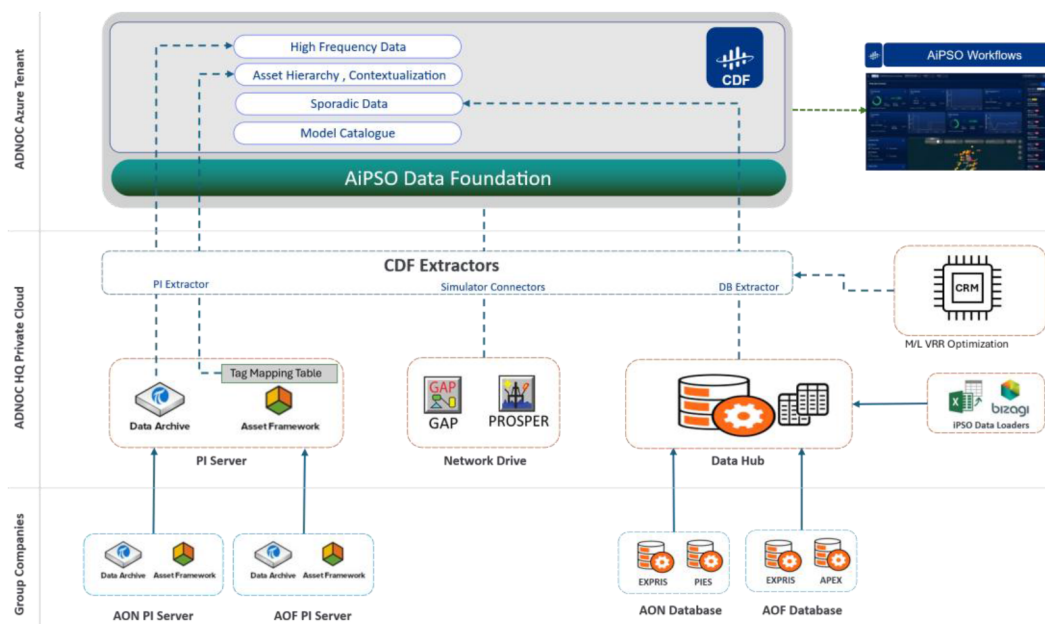


Figure 2—Production Ecosystem Orchestration Diagram

### Innovate Core Engineering Workflows Deployed

Core engineering workflows are the structured, repeatable processes that transform engineering inputs such as sensor data, simulation results, and field measurements into validated outputs that drive operational decisions. These workflows integrate domain expertise with standardized engineering practices, combining modeling and simulation. They typically involve iterative feedback loops, where results from one stage are validated, refined, and fed back into earlier steps to improve accuracy and reliability. Orchestrated through digital platforms and engineering toolchains, core workflows ensure traceability, compliance with industry standards, and alignment with project objectives, while enabling seamless collaboration across multidisciplinary teams.

**Electro Submersible Pump (ESP) Diagnostics.** ESP diagnostics continuously monitor key performance parameters such as pump intake pressure, motor current, vibration levels, and calculated pump efficiency to provide early detection of performance deviations and potential failure. By analyzing trends and comparing real-time data against historical baselines and manufacturer curves, the system can identify signs of pump degradation, scaling, gas interference, or blockage. Advanced analytics highlight anomalies that may indicate mechanical wear or operational imbalance (Villa et al, 2017). These insights enable proactive recommendations for targeted investigations, operational adjustments, or controlled rerun scenarios, allowing intervention before full failure occurs and minimizing unplanned downtime while extending equipment life (Figure 3).

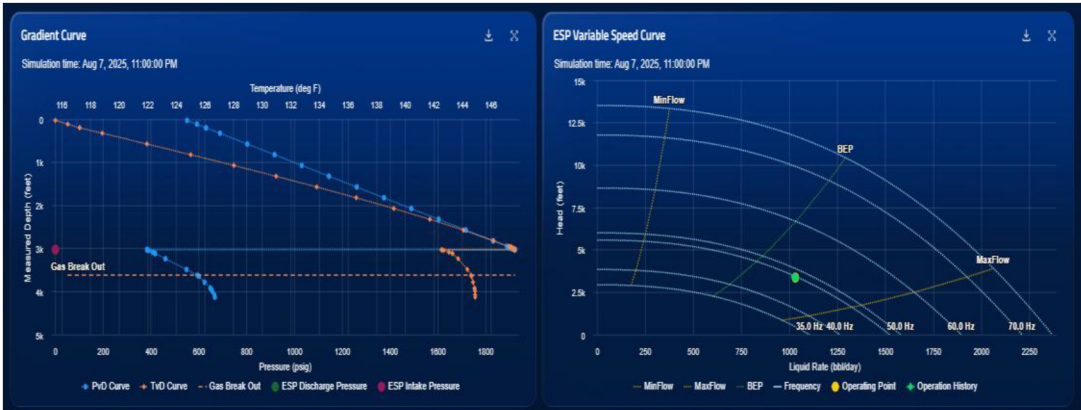


Figure 3—ESP Performance and Tornado Chart

**Gas Lift Diagnostics.** Gas lift diagnostics continuously monitor key operational parameters like injection gas rates, tubing and casing pressures, and well performance metrics to evaluate the efficiency and effectiveness of artificial lift operations. By analyzing trends and deviations from expected behavior, the system can detect issues such as multipointing, or injection inefficiencies that reduce lift performance. It also identifies suboptimal gas utilization, ensuring that the injected gas volume aligns with production requirements without being over or under injection (Misra, et al, 2024). By comparing real-time data to optimal lift curves derived from well models and historical performance, gas lift diagnostics provide actionable insights that enable operators to adjust injection strategies, maintain production stability, and maximize hydrocarbon recovery while minimizing operational costs. (Figure 4)

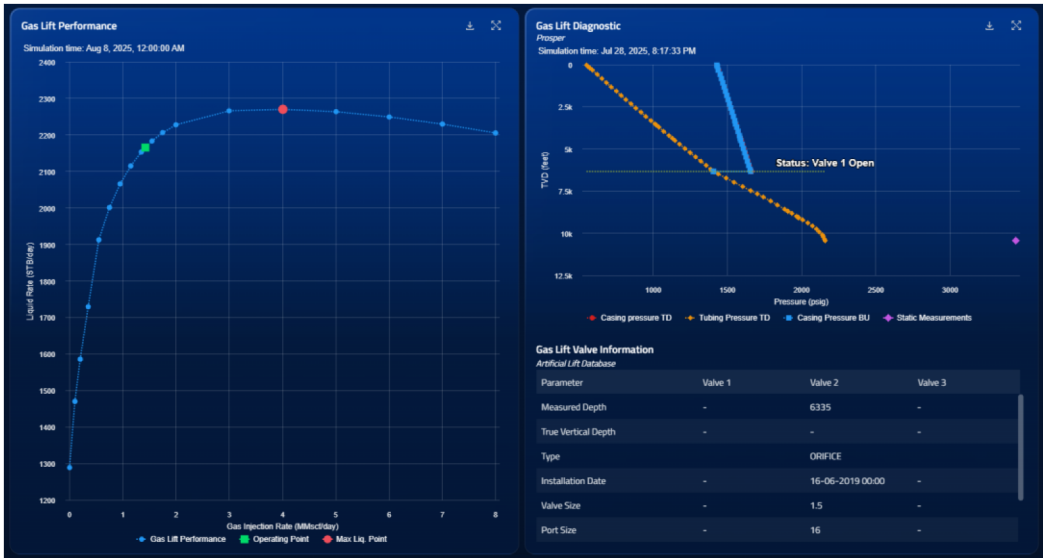


Figure 4—Gas Lift Performance and Diagnostics

**Volume Replacement Ratio & Water Injection Optimization (ML/Physics and CRM workflow).** This in-house solution is embedded into the AiPSO production ecosystem, leveraging a pre-developed Machine Learning (ML) solution in order to provide a rapid, robust, and reliable framework for optimizing production and injection rates in waterflooding project and gas injection operations (D. Badmaev et al, 2021). The primary objective is to maximize hydrocarbon recovery while ensuring the stability of the Voidage Replacement Ratio (VRR) for each injection pattern in the reservoir. The system employs the Capacitance Resistance Model (CRM) that is a data-driven (DD) methodology, integrating real-time field production data with predefined operational constraints and production quotas / targets. The embedded ML model processes



this information to recommend optimal injection rates. This predictive and prescriptive capability ensures effective reservoir pressure maintenance and sustains an optimal VRR at the individual pattern level.

**Network Optimization & Scenario planning.** An advanced and complete digital twin model of the entire production system was developed by integrating pre-calibrated PROSPER single well models with a comprehensive GAP network. This integration is orchestrated and automated within the AiPSO production ecosystem, eliminating the need for manual updates and tedious processes.

The calibrated individual well models capture detailed inflow and outflow dynamics using real-time data. These models are incorporated into the network simulator, which provides a high-fidelity, equation-based solution for steady-state multiphase networks. The model handles pressure, flow, and temperature distributions across wells, chokes, pipelines, and surface facilities, delivering robust network optimization capabilities. This solution enables proactive creation and planning of production optimization scenarios, avoiding production deferment. Additionally, production targets (quotas) are used as inputs for "what-if" scenarios to ensure the network total production remains close to the desired objective (Figure 5) These optimization scenarios generate new set points for individual wells. The system automatically creates trackable opportunities, and the wells requiring new set points (well movements) are listed in the AiPSO Opportunity and Value Tracking.

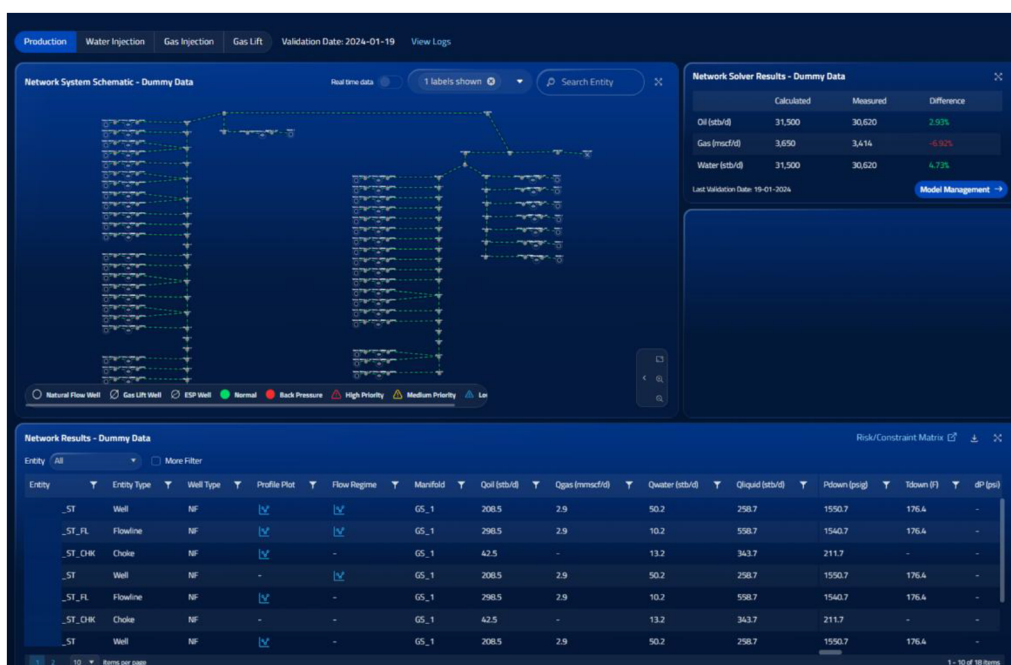


Figure 5—Network Optimization & Scenario planning

**Well test Validation and Path to surface normalization.** Surface well testing is a crucial process in the oil and gas industry that involves flowing fluids from a well to the surface to measure and evaluate its production potential and fluid characteristics (M. Hassan, 2006). The data collected from these tests provides essential information for making informed decisions throughout the life of a well. The end-to-end process of a surface well test begins with the controlled flow of hydrocarbons from the wellbore, where the multiphase fluid mixture (hydrocarbon) is directed through a specialized test unit, such as a multiphase flow meter (MPFM) or a test separator (TS). This unit is designed to continuously measure and monitor the individual flow rates of oil, gas, and water.

Following this real-time measurements, the raw data is subjected to a rigorous validation process to ensure its accuracy and reliability. The measured volumes are then normalized to standard conditions (SC), which allows for consistent comparison and reporting of production rates (M. Hoda and J. D. Asarpota, 2022).

Finally, the validated and normalized well test results are used to calibrate the well model, providing an accurate representation of the well's performance under current reservoir conditions. This iterative process of testing, validation, and model calibration is fundamental for effective reservoir management and production optimization, ensuring that operational decisions are based on the most current and reliable data available (Figure 6).

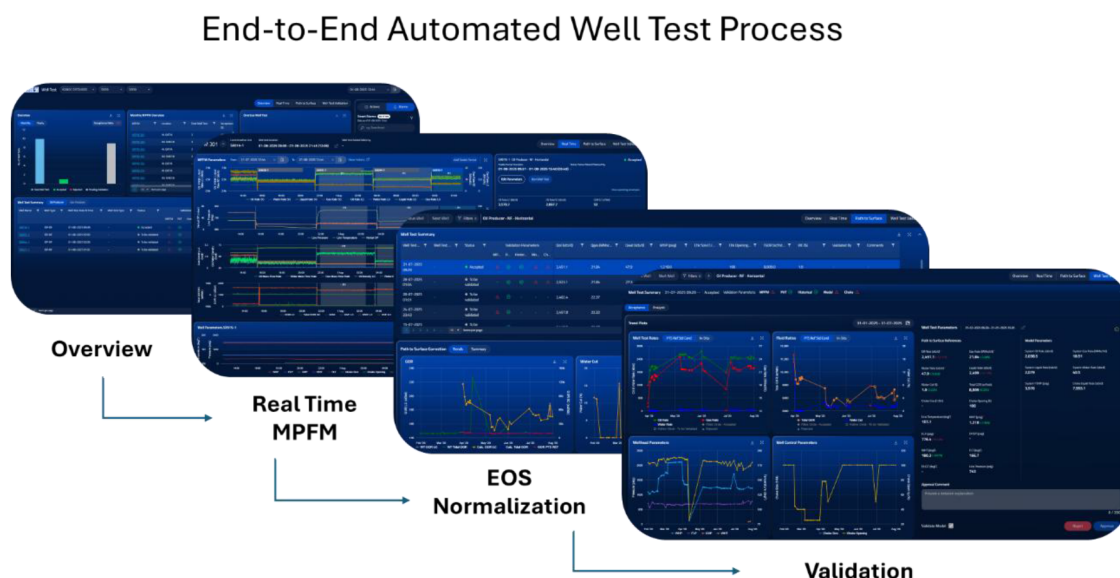


Figure 6—End to End Automated Well Test Process

**Automated Well and Network Model Management.** In the automated well and network model management system, each well model is continuously updated and calibrated with real-time field data. This process ensures the models accurately reflect current conditions, keeping them "evergreen." Once a well model is updated, it is automatically synchronized with the network model. This integrated approach guarantees that the entire production system consistently aligns with real production data, providing a reliable and up-to-date digital twin. All new physics models developed in this process are systematically stored within the AiPSO production ecosystem for future proactive decision making (Figure. 7)

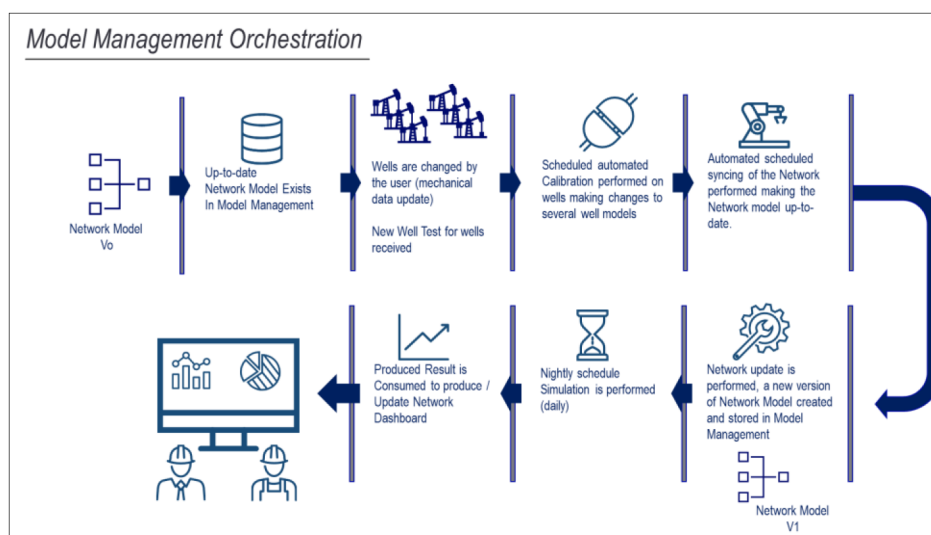


Figure 7—Automated AiPSO well and Network Model Management

**VFM-Based Rate Estimation.** VFM-based rate estimation leverages virtual flow metering models to infer wellhead flowrates in scenarios where multiphase flow meters (MPFMs) are limited, unavailable, or producing unreliable data. These models use calibrated pumps and well performance curves, along with pressure, temperature, and other operational field data, to estimate oil, gas, and water rates at wellhead conditions. By comparing the estimated flowrates with expected production envelopes and historical trends, operators can detect deviations, validate well performance, and identify potential issues such as flow restrictions or equipment underperformance. This approach provides critical redundancy in flowrate monitoring, ensuring reliable production insights even when physical metering is compromised, and supports informed operational decisions without interrupting production.

**Exception-based ticketing and notification.** Exception-based ticketing automates the identification, notification, and management of operational anomalies by generating alarms and generating tickets whenever predefined rules or thresholds are violated. Each notification captures critical context, including sparkline trends, operating envelope overlays, and flagged rate changes, enabling engineers to quickly assess deviations from normal well performance. The tickets are routed and prioritized on a centralized dashboard, sorted by discipline, severity, and urgency, ensuring that the right personnel receive actionable alerts in real time. This system streamlines issue tracking, reduces response time, and supports proactive decision-making by highlighting the most critical exceptions before they escalate into significant operational problems.

**Value Tracking.** Value tracking systematically quantifies the operational impact of resolved tickets by embedding metadata fields such as production deferred, downtime avoided, and remedial actions executed. Dashboards provide a comprehensive view of active, resolved, and recurring issues across assets, enabling engineers and management to monitor performance trends and prioritize interventions. KPI scorecards further consolidate this information, reporting the cumulative value of resolved tickets on a quarterly basis, which facilitates performance benchmarking, continuous improvement, and strategic decision-making. By linking operational actions to measurable outcomes, value tracking ensures that engineering efforts are aligned with production efficiency and business objectives.

All these capabilities are seamlessly integrated within a single platform. This unified system centralizes real-time monitoring, anomaly detection, automated alerting, and impact measurement, providing engineers with a holistic view of well performance and operational efficiency. By combining diagnostics, automated workflows, and actionable insights in one interface, the platform enables proactive decision-making, reduces downtime, optimizes production, and directly links interventions to measurable value, all without switching between multiple tools or data sources (Figure 8).





Figure 8—Well Surveillance by Exception Based using Visual Analytic

## Operational Impact and Strategic Significance

The production digital ecosystem AiPSO deployment across high-well-count assets covering over 500 wells demonstrated significant operational improvements.

- **Detection Latency:** Up to 80% faster identification of underperformance compared to manual reviews
- **Alarm Reduction:** Over 60% reduction in non-actionable alerts in pilot areas
- **Production Recovery:** Estimated 1–2% uplift in production via early interventions
- **Response Time:** Engineering actions initiated within hours vs. days previously
- **Cross-Team Coordination:** Standardized ticketing enabled seamless collaboration among production, reservoir, and lift engineers.

Beyond technical performance, the well surveillance initiative demonstrates a scalable model for transforming operational culture. By structuring workflows around exception logic and tying them to business value, field teams shift from reactive monitoring to proactive optimization. The integration of diagnostics, ticketing, and value tracking ensures accountability and encourages intervention impact measurement. This effort is aligned with AiPSO broader vision: embedding digital workflows that empower engineers, automate routine surveillance, and close the loop between detection and field action. As assets grow more complex, scalable surveillance becomes a competitive differentiator.

## Novelty and Contributions

This work demonstrates a novel integration of surveillance data, engineering logic, and cross-functional workflows to solve real-world field challenges. Contributions include:

- A modular, workflow-driven diagnostic framework for ESP and gas lift systems
- Embedded value-tracking and intervention analytics
- Scalable implementation aligned with ADNOC's operational structure

The system sets a new benchmark for well surveillance in mature, high-well-count assets and shows how structured engineering logic can drive tangible performance improvements in upstream operations.

## Conclusions

The AiPSO Well Surveillance and Diagnostics system provides a replicable blueprint for digital transformation in high-well-count environments. By combining structured engineering workflows, calibrated models, and smart ticketing, the system delivers measurable improvements in well performance monitoring and intervention effectiveness.

The end-to-end automated process for surface well testing provides the essential data needed to build accurate well models, which are then used to inform strategic decisions for reservoir management. This process, from measurement to normalization and calibration, ensures that production optimization efforts are based on the most reliable and up-to-date information available.

The advanced digital twin system connects individual well models with a complete network simulator, automating production optimization. It allows engineers to test "what-if" scenarios, ensuring production targets are met. The system provides actionable insights to maximize asset value and operational efficiency.

The in-house CRM data-driven methodology solution leverages ML to optimize injection rates, maximizing hydrocarbon recovery while maintaining reservoir health. By integrating real-time data with production operational targets, the system provides proactive and precise recommendations. Ultimately, this ensures effective pressure maintenance and sustains an optimal VRR at the reservoir pattern level.

Value tracking links operational set points directly to business results. By quantifying the impact of resolved issues or opportunities like downtime avoided or production gained on a single platform. This method ensures all efforts are focused on improving efficiency and meeting strategic ADNOC goals.

This initiative proves that operational excellence can be scaled through automation and engineering discipline, enabling field teams to shift from passive monitoring to value-based decision-making. As deployment continues across ADNOC assets, the system is evolving into a continuous assurance platform, driving production reliability and cost efficiency.

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